Robust Acoustic Speaker Tracking & Multimedia Presentation Systems

Live Benchmark Deck with Beamer Notes, Video, and Audio

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PRESENTER NOTES (Slide 1):

- Welcome everyone to the presentation.
- Introduce the objective: evaluating the next-generation Rust-based Beamer Presenter.
- Point out that this presentation includes embedded audio and video media.
- Press ‘L‘ on your keyboard to activate the laser pointer at any time.

Mathematical Modeling: Acoustic State Space

State-Space Equation for Moving Acoustic Source

Let $\mathbf{x}_k = [p_x, p_y, \dot{p}_x, \dot{p}_y]^T$ denote the state vector of the active speaker:

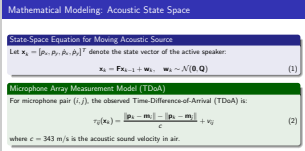
$$\mathbf{x}_k = \mathbf{F}\mathbf{x}_{k-1} + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \tag{1}$$

Microphone Array Measurement Model (TDoA)

For microphone pair (i, j), the observed Time-Difference-of-Arrival (TDoA) is:

$$\tau_{ij}(\mathbf{x}_k) = \frac{\|\mathbf{p}_k - \mathbf{m}_i\| - \|\mathbf{p}_k - \mathbf{m}_j\|}{c} + v_{ij} \tag{2}$$

where $c = 343$ m/s is the acoustic sound velocity in air.



SPEAKER NOTES (Slide 2):

- Walk the audience through the continuous-to-discrete transition matrix **F**.
- Highlight the non-linear hyperbolic nature of the TDoA measurements.
- If needed, use the **Pen tool (P)** to sketch the microphone array baseline.
- Or press '+' or 'N' to open a blank whiteboard for step-by-step Taylor expansion.

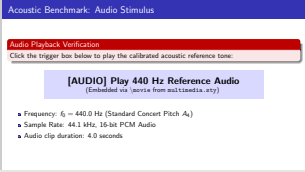
Acoustic Benchmark: Audio Stimulus

Audio Playback Verification

Click the trigger box below to play the calibrated acoustic reference tone:

[AUDIO] Play 440 Hz Reference Audio
(Embedded via \movie from multimedia.sty)

- Frequency: $f_0 = 440.0$ Hz (Standard Concert Pitch A_4)
- Sample Rate: 44.1 kHz, 16-bit PCM Audio
- Audio clip duration: 4.0 seconds



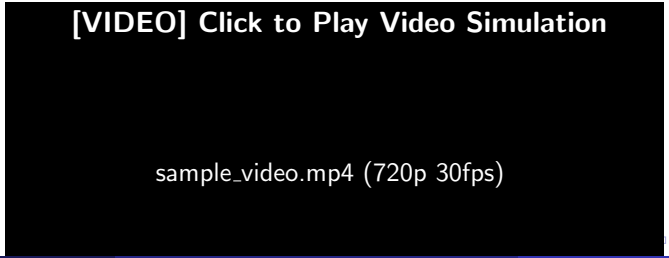
SPEAKER NOTES (Slide 3):

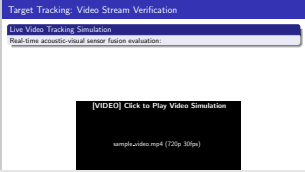
- Play the 440 Hz test tone to demonstrate native audio subsystem integration.
- Confirm that audio plays through the main presentation audio output without lag.
- Explain that speaker tracking algorithms initialize spectral baselines with this reference.

Target Tracking: Video Stream Verification

Live Video Tracking Simulation

Real-time acoustic-visual sensor fusion evaluation:





SPEAKER NOTES (Slide 4):

- Start the video playback.
- Note the zero-tearing and 30fps smooth hardware-accelerated decode.
- Highlight the test color patterns simulating camera sensor input.
- Mention that the presenter scrub bar allows seeking to any timecode seamlessly.

Summary & Live Inking Demonstration

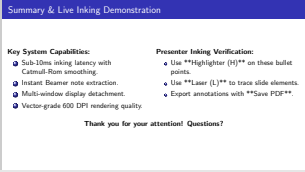
Key System Capabilities:

1. Sub-10ms inking latency with Catmull-Rom smoothing.
2. Instant Beamer note extraction.
3. Multi-window display detachment.
4. Vector-grade 600 DPI rendering quality.

Presenter Inking Verification:

- Use **Highlighter (H)** on these bullet points.
- Use **Laser (L)** to trace slide elements.
- Export annotations with **Save PDF**.

Thank you for your attention! Questions?



SPEAKER NOTES (Slide 5):

- Open the floor for questions from committee or conference audience.
- Show how easy it is to draw derivations live on the slides or on whiteboard pages.
- Export the final annotated presentation to PDF when the session concludes.